

Simulation-Based Insights into the Evolution of Mammalian Search Behaviour

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1 How do animals navigate their world?

To survive and reproduce, animals must balance positive and negative encounters^{1,2}

Positive Encounters

- food
- shelter
- mates

Negative Encounters

- predation
- disease
- competition

Ballistic movement (targeted) → *Increases* all encounters

Diffusive movement (tortuous) → *Decreases* all encounters

Therefore, animals who face increased **risk** should **move more diffusively**, even if it means finding *less food*^{1,3,4}

2 Movement simulations

input parameters



prey features

- mass
- number of individuals

habitat
(food raster)



number of generations and repetitions

movement simulation (continuous time movement model)

movement parameters

If generation 1:
body mass scaling
Later generations:
inherited values

foraging

patch detection via spatial overlay

resource allocation

resources are allocated to metabolic and movement costs, growth, and reproduction

simulate movement



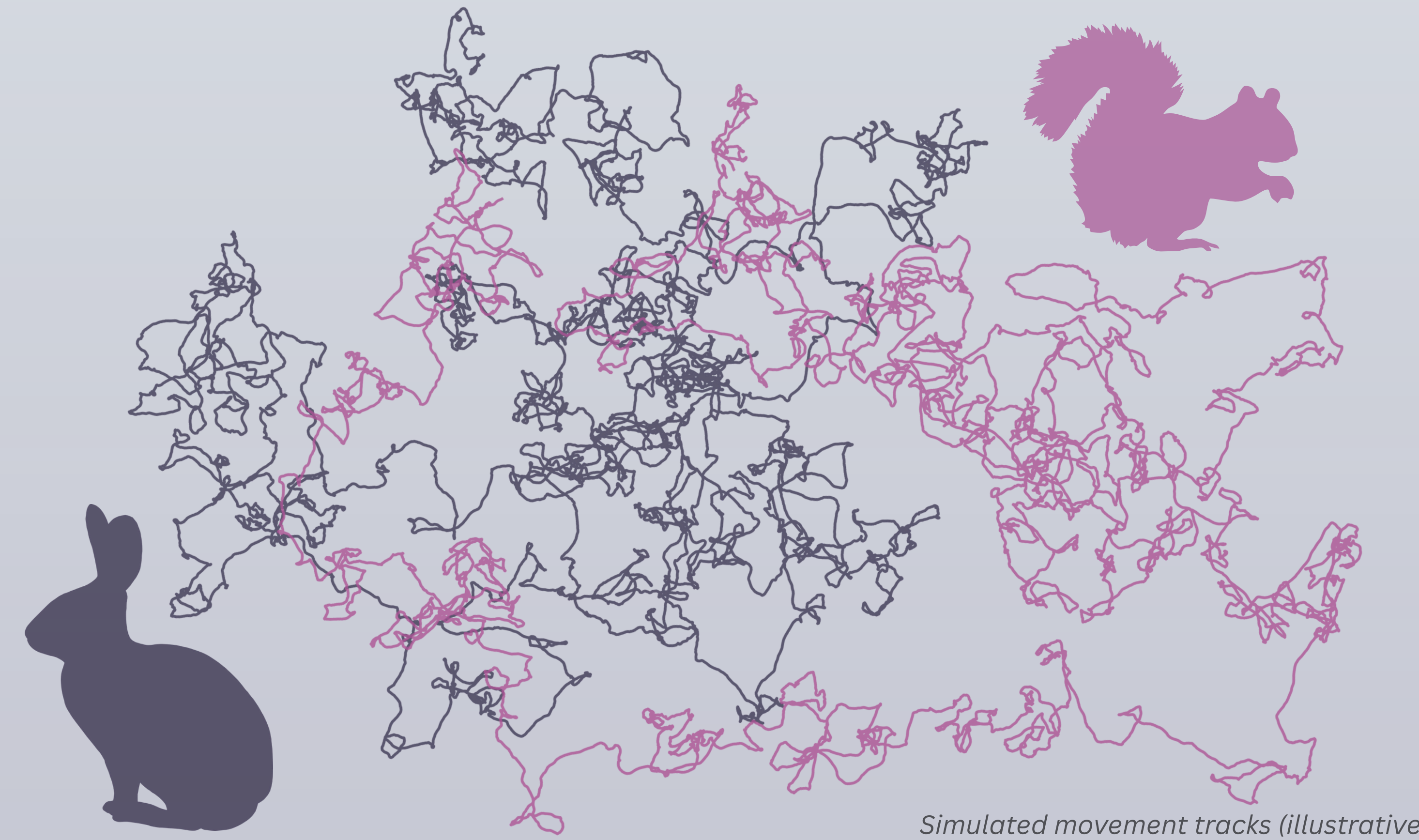
output parameters

ballistic length scale (m)

reproduction success

evolved traits

We simulated **multi-generational movement** under **constant environmental conditions** to explore how search strategies evolve in prey species



Simulated movement tracks (illustrative)

3 Energetics shape movement

Energetic costs influence the optimal movement strategy

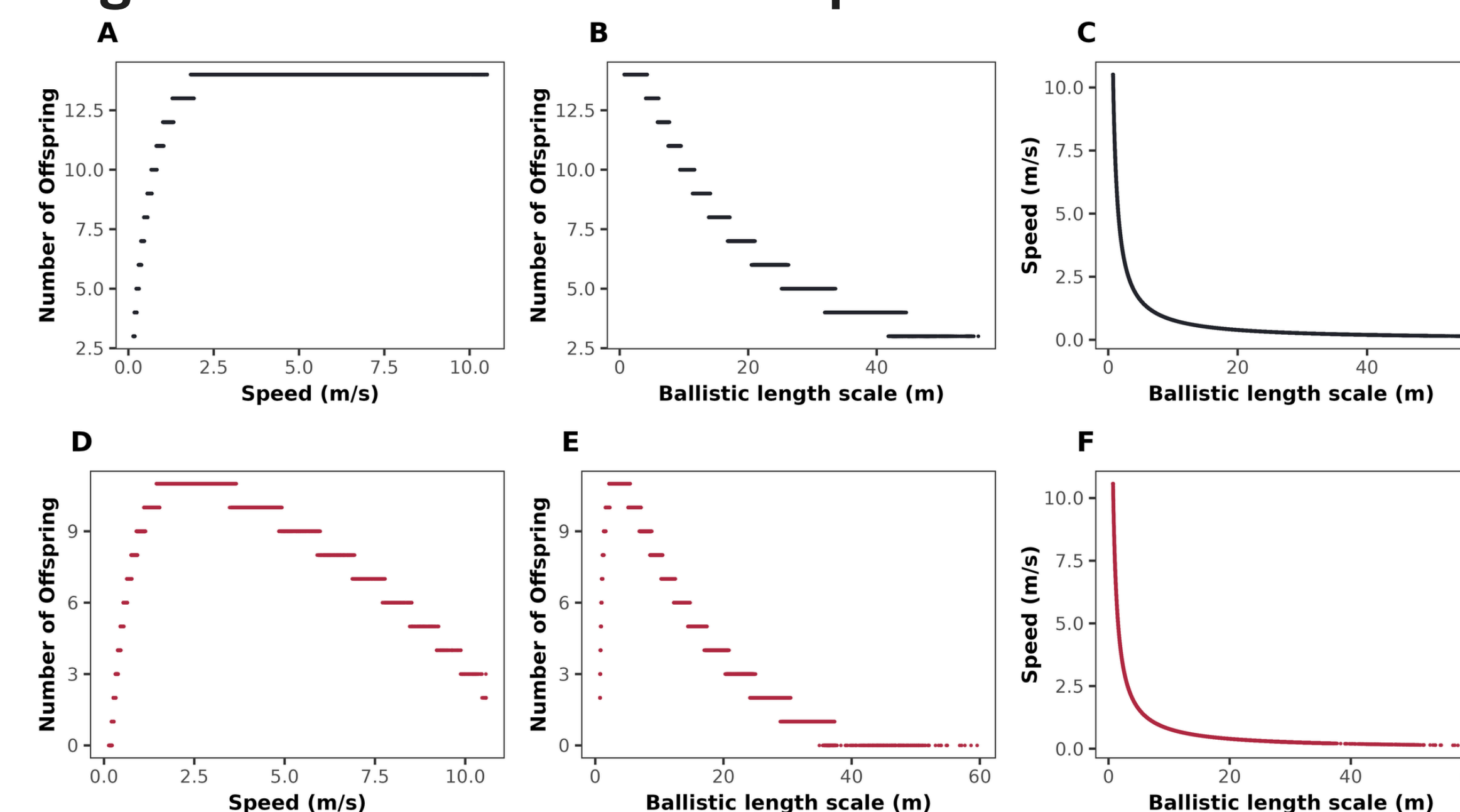


Figure 1: Supplementary analysis for a 105.5 kg individual. Panels A-C show how speed, ballistic length scale (l_v), and net calories interact without energetic costs; Panel D shows predicted net calories from l_v and speed. Panels E-G show the same with energetic costs; Panel H shows the corresponding predictions. Energetic costs lower net calories and select for an optimum combination of speed and l_v .

Stable movement strategies emerge

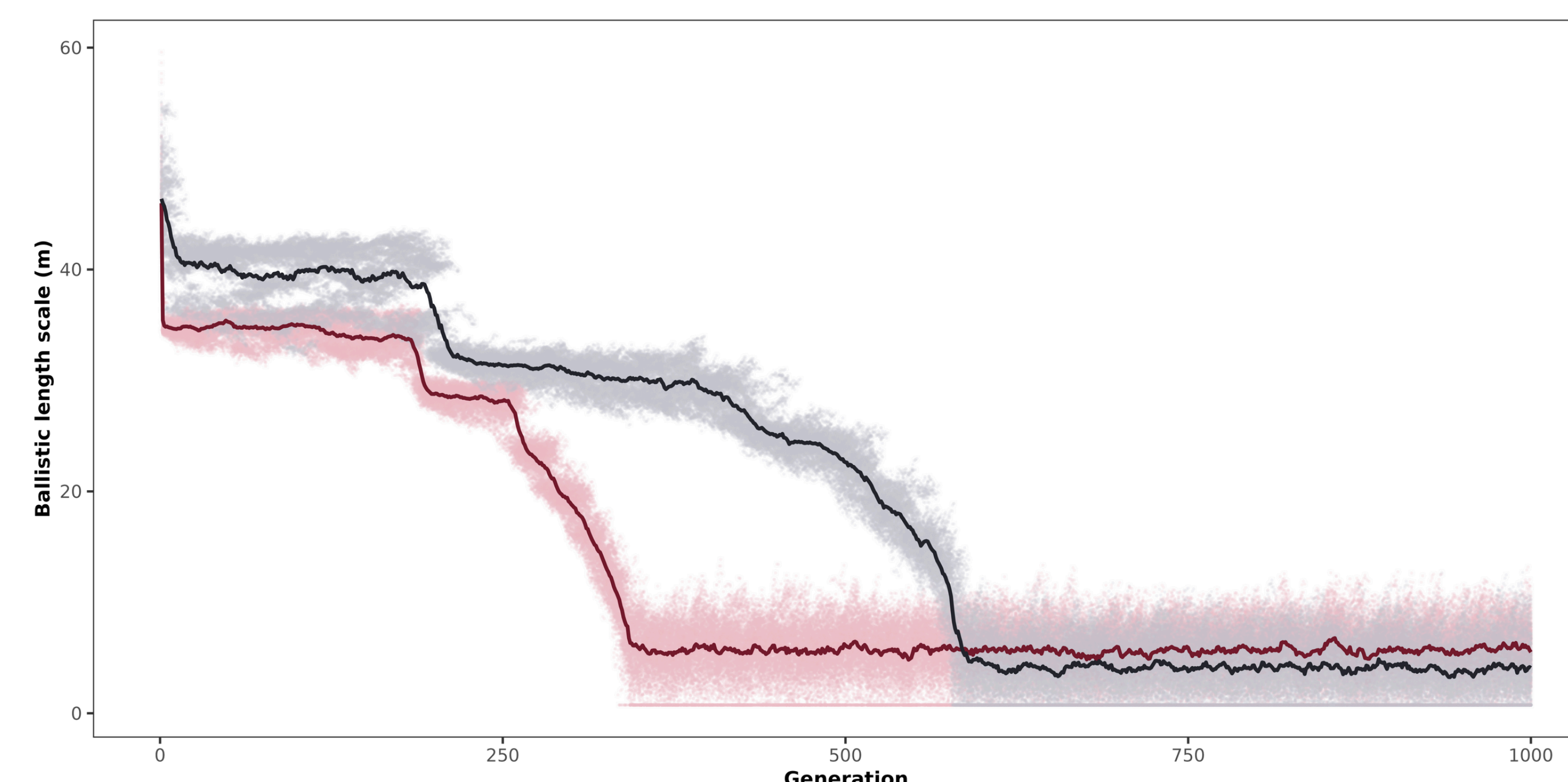


Figure 2: Evolution of ballistic length scale (l_v) over 1000 generations for a 105.5 kg individual, without energetic costs (black) and with energetic costs (red). Energetic costs influence the rate of evolution.

4 How do search strategies scale?

Ballistic length scale increases nonlinearly with body size

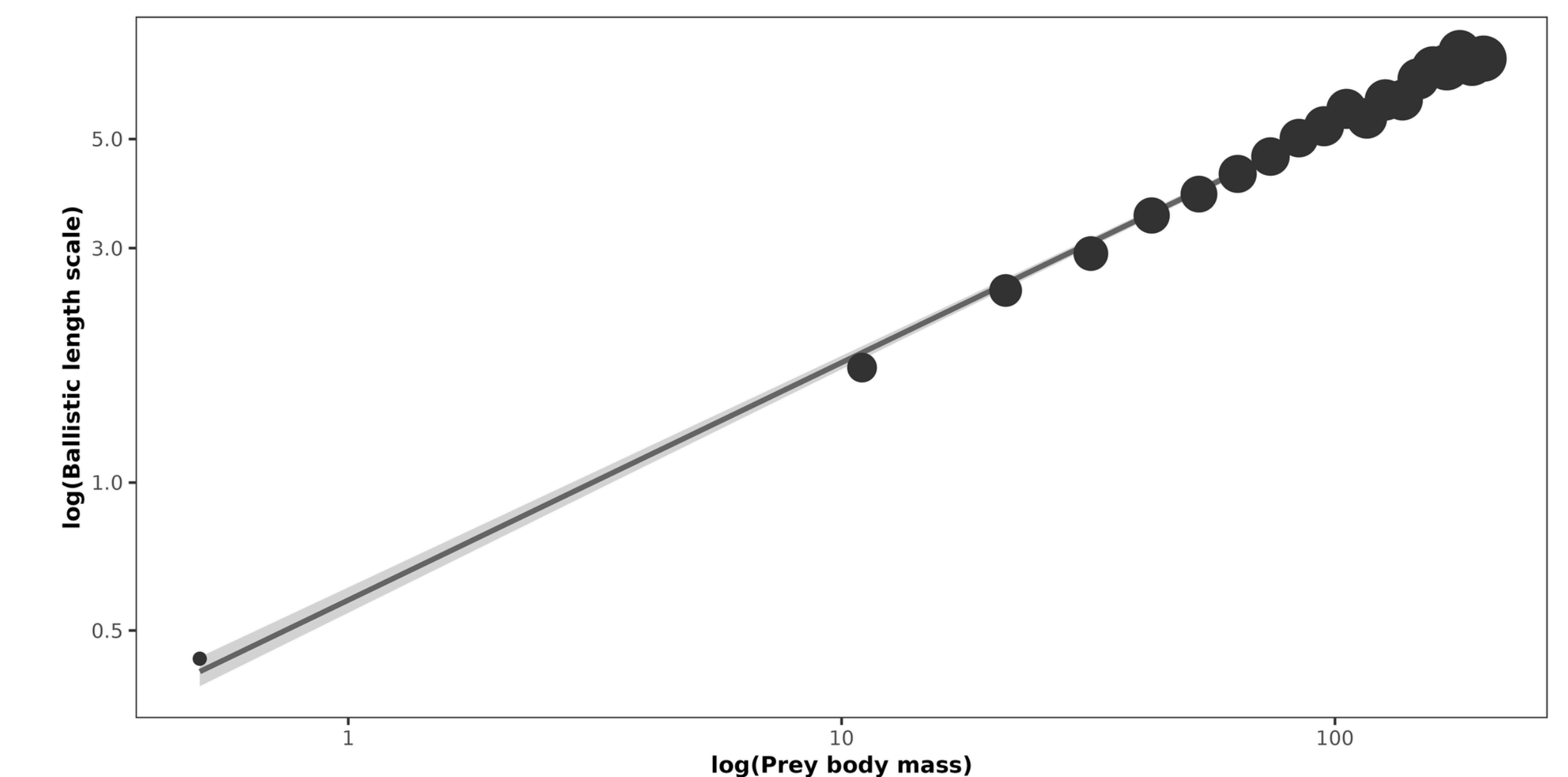


Figure 3: Relationship between prey body mass (kg) and ballistic length scale (l_v). Panel A shows the mean l_v across the final 100 generations (point size indicating the standard error), with a smoothed fit and 95% confidence interval. Panels (i) and (ii) show example movement tracks for a 42.5 kg individual with low l_v (i) and a 200 kg individual with a higher l_v (ii). Ballistic length scale increases nonlinearly with body size, producing more directed movement at higher masses.

Nonlinear effects drive deviations at mass extremes

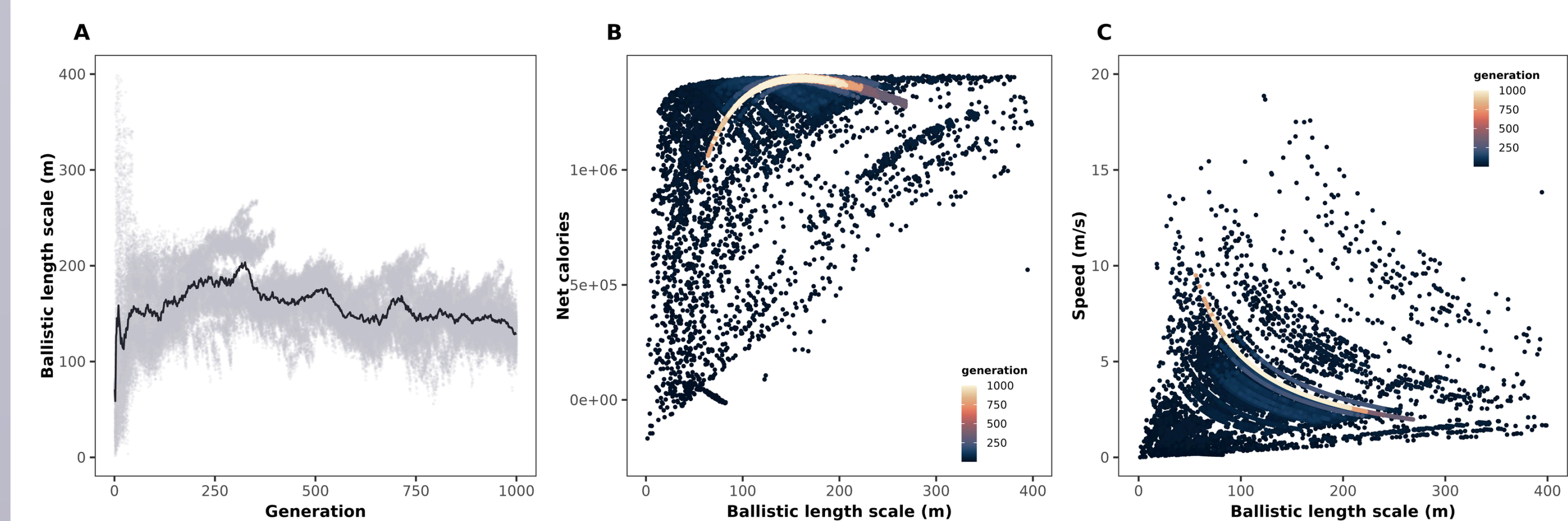


Figure 4: Simulation results for a 200 kg individual. Panel A shows the change in ballistic length scale (l_v) over 1000 generations (the black line is the population mean, grey points are individual values). Panel B shows net calories versus l_v , and Panel C shows speed versus l_v , both are coloured by generation. Extremes in body mass showed departures from the clean pattern observed at mid-range values (see figure 1).

5 Next steps

- Complete simulations for all prey body sizes in constant, predator-free environments
- Add predators to explore how top down pressure shapes movement strategies

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